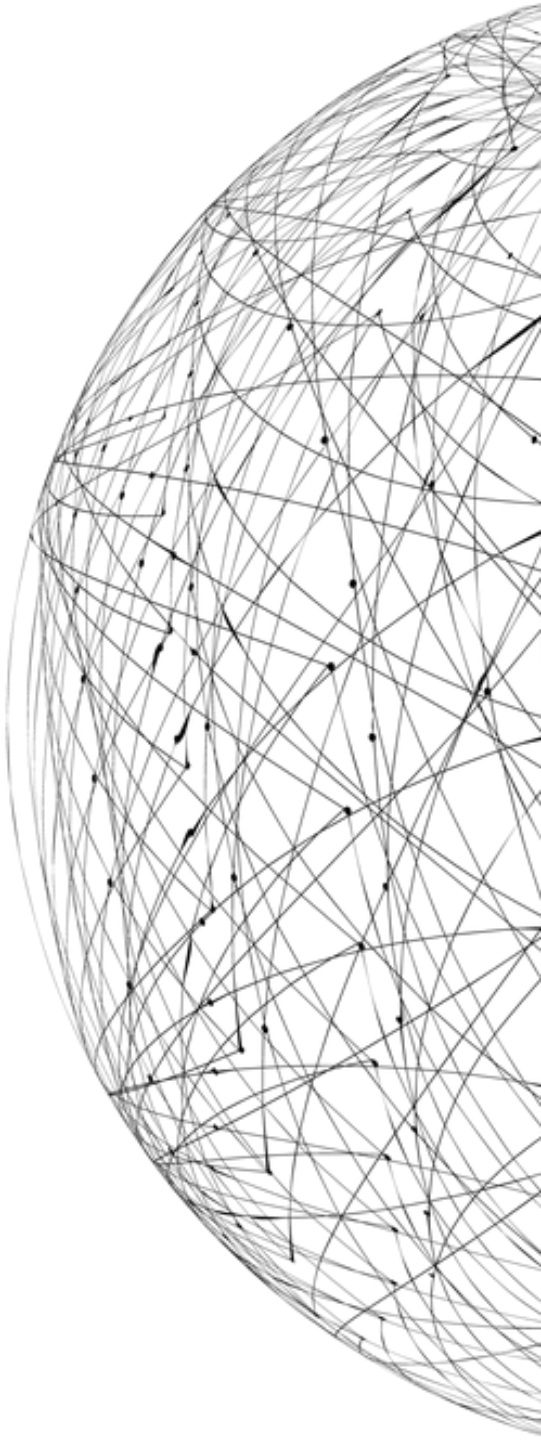
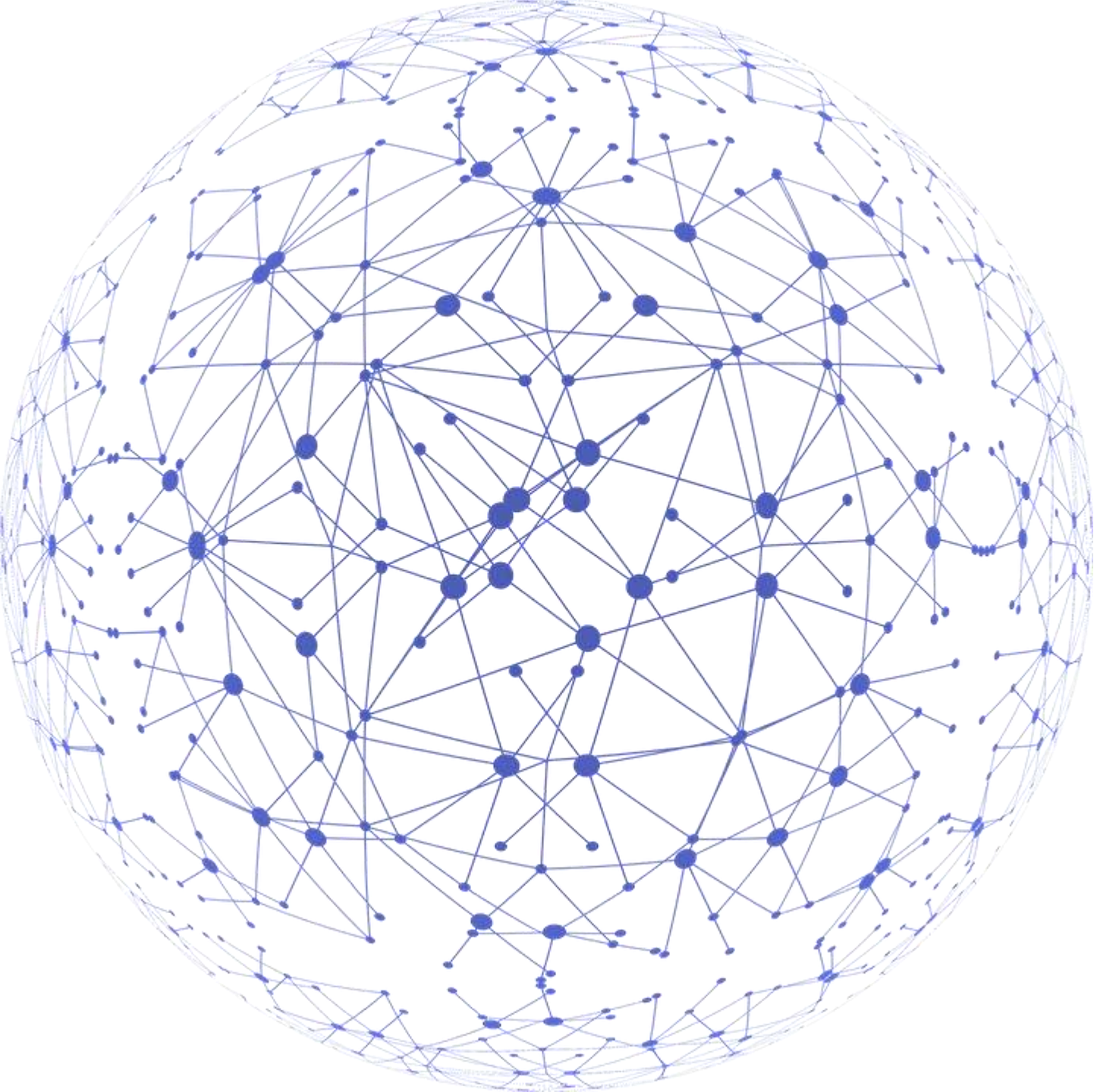
Internet Programming

Chapter 1: Introduction





Content

- Introduction and History
- Internet Services
- Types of Web Pages
- Client-Side Scripting
- Server-Side Scripting

Introduction

- What is the **Internet**?

Introduction (cont…)

- The **internet** is a global computer network providing a variety of information and communication facilities, consisting of interconnected networks using standardized communication protocols.

Oxford
Dictionaries

Introduction (cont…)

- The **internet** is a globally connected network system that uses TCP/IP to transmit data via various types of media. The internet is a network of global exchanges – including private, public, business, academic and government networks – connected by guided, wireless and fiber-optic technologies.

techopedia™

Introduction (cont…)

- WAN is the generic for any Wide Area Network. And the Internet is a particular implementation of a WAN, using TCP/IP packets and supports http protocol.
- Computers connected to a wide-area network are often connected through public networks, such as the telephone system. They can also be connected through leased lines or satellites. The largest WAN in existence is the Internet.

Introduction (cont…)

- Internet:
  - Is an international network of networks that links together millions of computers.
  - Is commonly referred to as the 'Net’.
  - Is a “network of networks” that consists of smaller domestic, academic, business, and government networks.
  - Is not a closely controlled ‘single’ system, rather it is an assemblage of independent networks united by the common use of TCP/IP protocol.

History

- Overview
  - **1960s**: The precursor to the internet, ARPANET, was created by the U.S. Department of Defense's Advanced Research Projects Agency (ARPA). It connected four major universities to facilitate research and communication.
  - **1970s**: ARPANET continued to expand, incorporating more universities and research institutions. This period saw the development of email as a means of communication over the network.
  - **1980s**: TCP/IP (Transmission Control Protocol/Internet Protocol) was standardized, allowing different types of computers and networks to communicate with each other, forming the basis of the modern internet. The National Science Foundation Network (NSFNET) was established, providing high-speed connectivity to academic and research institutions.

History

- Overview (cont...)
  - **1990s**: The World Wide Web was invented by Tim Berners-Lee at CERN in Switzerland. It introduced the concept of hyperlinks and web pages, making information accessible and navigable through a graphical interface.
  - **2000s**: The dot-com bubble burst in the early 2000s, leading to a period of consolidation and restructuring in the internet industry. Social media platforms like Facebook and Twitter emerged, revolutionizing online communication and social networking. Broadband internet became widely available, allowing for faster and more reliable internet connections.

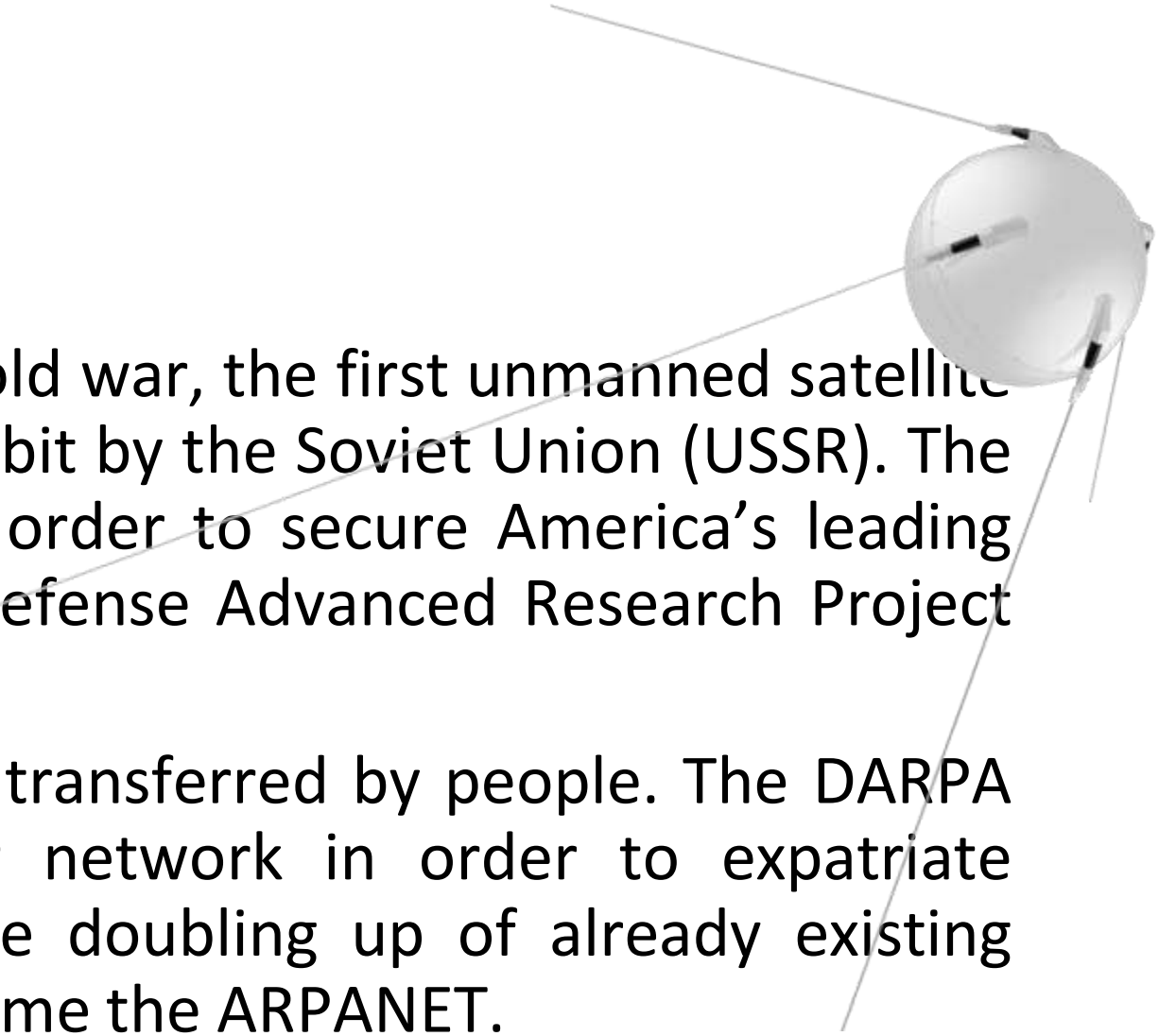
History

- Overview (cont...)
  - **2020s**: The internet continues to evolve with advancements in technologies like artificial intelligence, virtual reality, and blockchain. Issues surrounding online privacy, cybersecurity, and digital divide are increasingly prominent. The COVID-19 pandemic further highlighted the importance of the internet for remote work, education, and socialization.

History

- Before 1957, computers only worked on one task at a time. This is called Batch Processing, of course it was quite ineffective. ☹
- With computers getting bigger and bigger, they had to be stored in a special cooled rooms, but then the developers couldn’t work directly on the computers anymore. ☹
- The year 1957 marked a big change. A remote connection had to be installed, so that the developers could work directly on the computers. (at the same time the idea of time sharing came up) ☺
- This is the first concept in computer technology to share the processing power of one computer with multiple users.

History (cont…)



- On October, 4th 1957 during the cold war, the first unmanned satellite called “Sputnik 1” was sent into orbit by the Soviet Union (USSR). The fair of a missile gap emerged. In order to secure America’s leading technology the US founded the Defense Advanced Research Project Agency in February, 1958.
- At that time knowledge was only transferred by people. The DARPA planned a large scale computer network in order to expatriate knowledge transfer and avoid the doubling up of already existing research. This network would become the ARPANET.

History (cont…)

- Further more, three other concepts would to be developed, which are fundamental for the history of the internet:
  - The concept to the military network by the RAND corporation in America.
  - The commercial network of the National Physical Laboratory (NPL) in England.
  - The scientific network CYCLADES in France.
- The scientific, military and commercial approaches of these concepts are the foundations for our modern internet.

History (cont…)

- **ARPANET** (development begun in 1966)
  - Universities was generally quite cautious about sharing their computers, therefore small computers were put in front of the mainframe. This computer, the Interface Message Processor (IMP) took over control of the network activities, while the mainframe was only in charge of the initialization of programs and data files.
  - Since only the IMPs were interconnected in a network, this was also called IMP subnet.
  - For the first connections between the computers, the Network Working Group developed the Network Control Protocol (NCP). Letter on the NCP was replaced by the more efficient Transmission Control Protocol (TCP).
  - The specific feature of the TCP is the verification of the file transfer.

History (cont…)

- National Physical Laboratory (**NPL**)
  - Since the NPL network was designed on a commercial bases, a lot of users and file transfer were expected. In order to avoid congestion of the lines, the sent files were divided into smaller packets, which were put together again at the receiver. **PACKET-SWITCHING** was born.

History (cont…)

- In 1962, American ferret aircrafts discovered middle and long range missiles in Cuba, which were able to reach the United States. This stoked fear of an atomic conflict. At that time information systems had a centralized network architecture. To avoid breakdown during an attack, a decentralized network architecture had to be developed, which in case of loss of a node would still be operative.
- Communication still used to work through radio waves, that would have caused problems in case of atomic attack: the ionosphere would be affected and the long-wave radio waves wouldn’t work anymore. Therefore they had to use direct waves, which, however, don’t have a long range.
- A better solution was the model of a distributed network. Thus long distances could be covered with a minimum of interferences.

History (cont…)

- Another milestone followed with the development of the French network "**Cyclades**". Since Cyclades had a far smaller budget than ARPANET and thus also fewer nodes, the focus was laid on the communication with other networks. In this way the term "**inter-net**" was born.
- Moreover, Cyclades' concept went further than ARPA's and the NPL's. During communication between sender and receiver the computers went not to intervene anymore, but simply server as a transfer node. Cyclades' protocol went through all machines using a physical layer that was implemented into the hardware, providing a direct connection with the receiver – an end-to-end structure.

History (cont…)

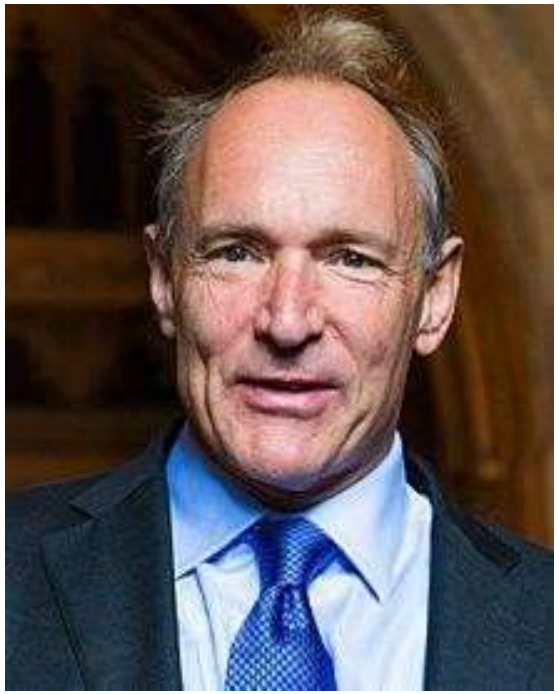
- Inspired by the Cyclades network and driven by the incompatibility between the networks, their connection gained in importance everywhere.
- The phone companies developed the X.25 protocol which enables communication through their servers – in exchange for monthly basic charge of course.
- DARPA’s Transmission Control Protocol (TCP) was to connect the computers through get-ways. And the International Organization for Standardization designed the OSI Reference Model. The innovation of OSI was the attempt to standardized the network from its ends, and the channel’s division into a separate layers.

History (cont…)

- Finally the TCP assimilated the preferences of the OSI Reference Model and gave way to the TCP/IP protocol – a standard which guaranteed compatibility between networks and finally merged them, creating the **Internet**.
- By February 28, 1990 the ARPANET hardware was removed, but the internet was up and running.

History (cont…)

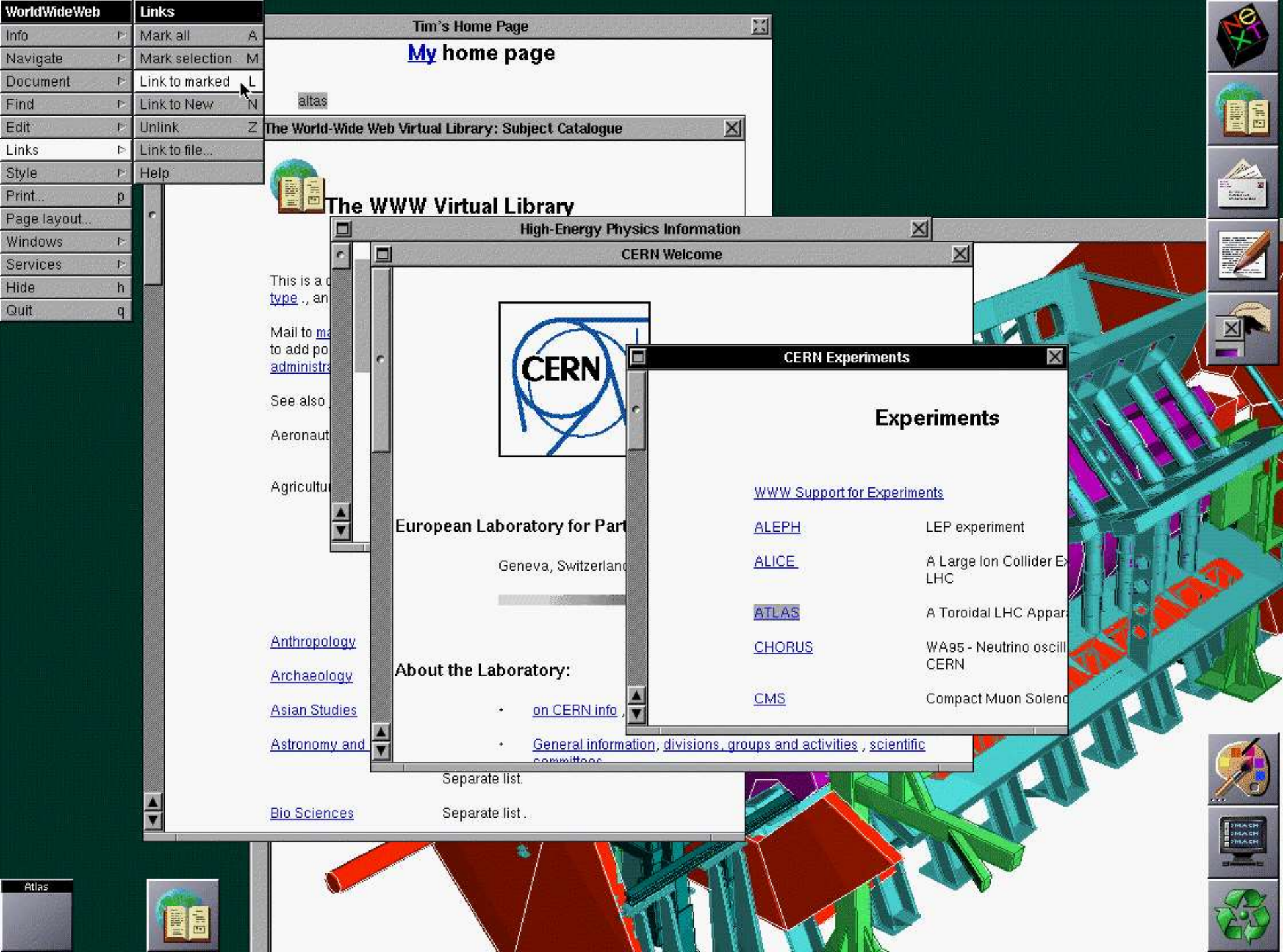
- The World Wide Web (WWW), also called the Web, is an information space where documents and other web resources are identified by Uniform Resource Locators (URLs), interlinked by hypertext links, and accessible via the Internet. English scientist **Tim Berners-Lee** invented the World Wide Web in **1989**. He wrote the first web browser in 1990 while employed at CERN in Switzerland.
- The browser was released outside CERN in 1991, first to other research institutions starting in January 1991 and to the general public on the Internet in August 1991.



Berners-Lee in 2014
(Source: Wikipedia)

History (cont…)

- The WorldWideWeb browser:
  - The first web browser, after all, when it was written in 1990 by **Tim Berners-Lee**, it was the only way to see the web. Much later it was renamed **Nexus** in order to save confusion between the program and the abstract information space (which is now spelled World Wide Web with spaces).
  - **Berners-Lee** wrote the program using a NeXT computer.



***WorldWideWeb* (Nexus)**

The first web browser

Source (w3.org)

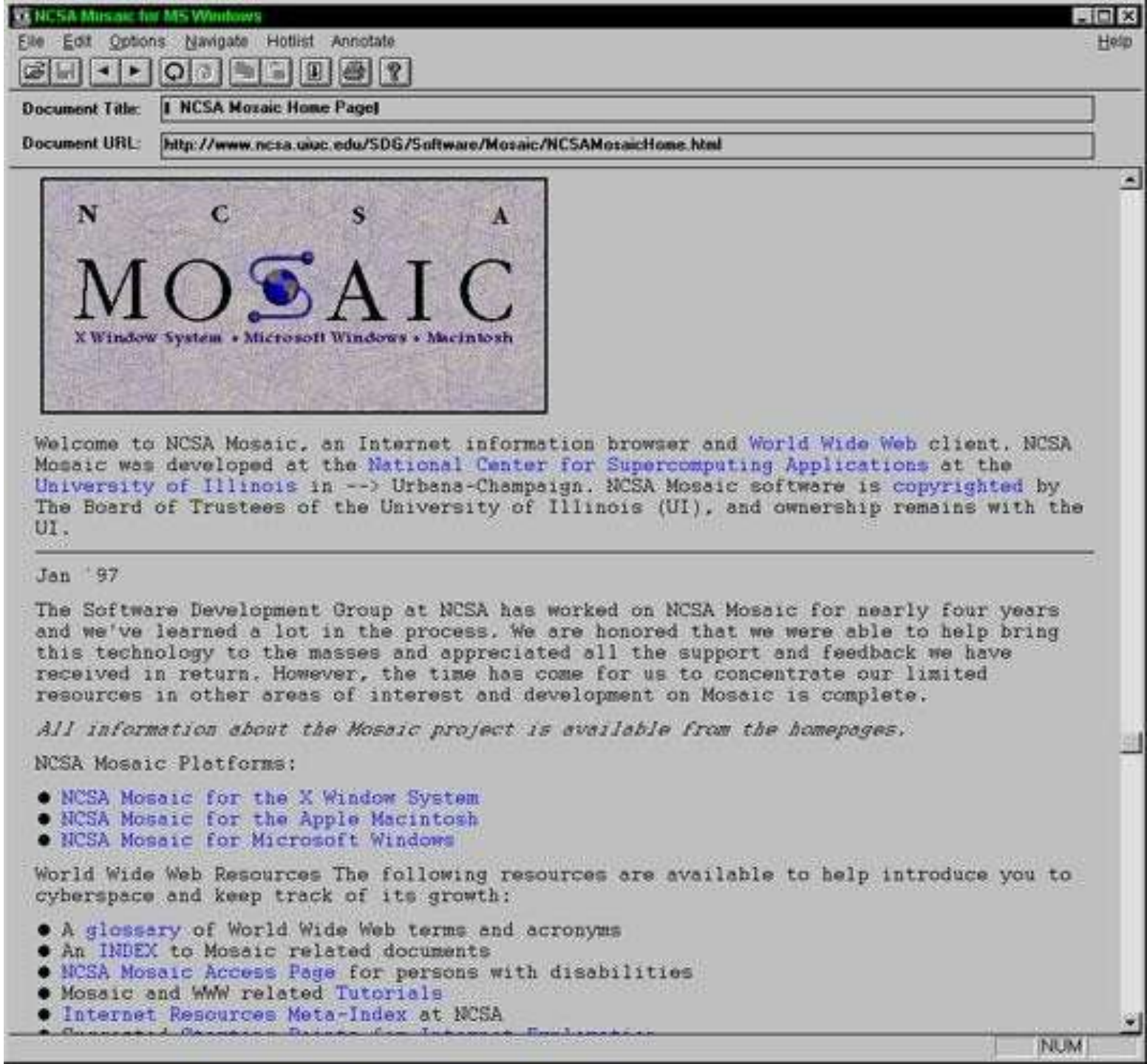
History (cont…)

- Mosaic (1992):
  - NCSA Mosaic was neither the first web browser (*first was the WorldWideWeb of Berners-Lee*) nor the first graphical web browser (*it was preceded by the lesser-known Erwise and ViolaWWW*), but it was the web browser credited with popularizing the World Wide Web.
  - Its clean, easily understood user interface, reliability, Windows port and simple installation all contributed to making it the application that opened up the Web to the general public.

Eric Bina (left) and Marc Andreessen (right)
Source (history-computer.com)

NCSA Mosaic Browser

Source (history-computer.com)



History (cont…)

- Today’s popular web browsers:

History (cont…)

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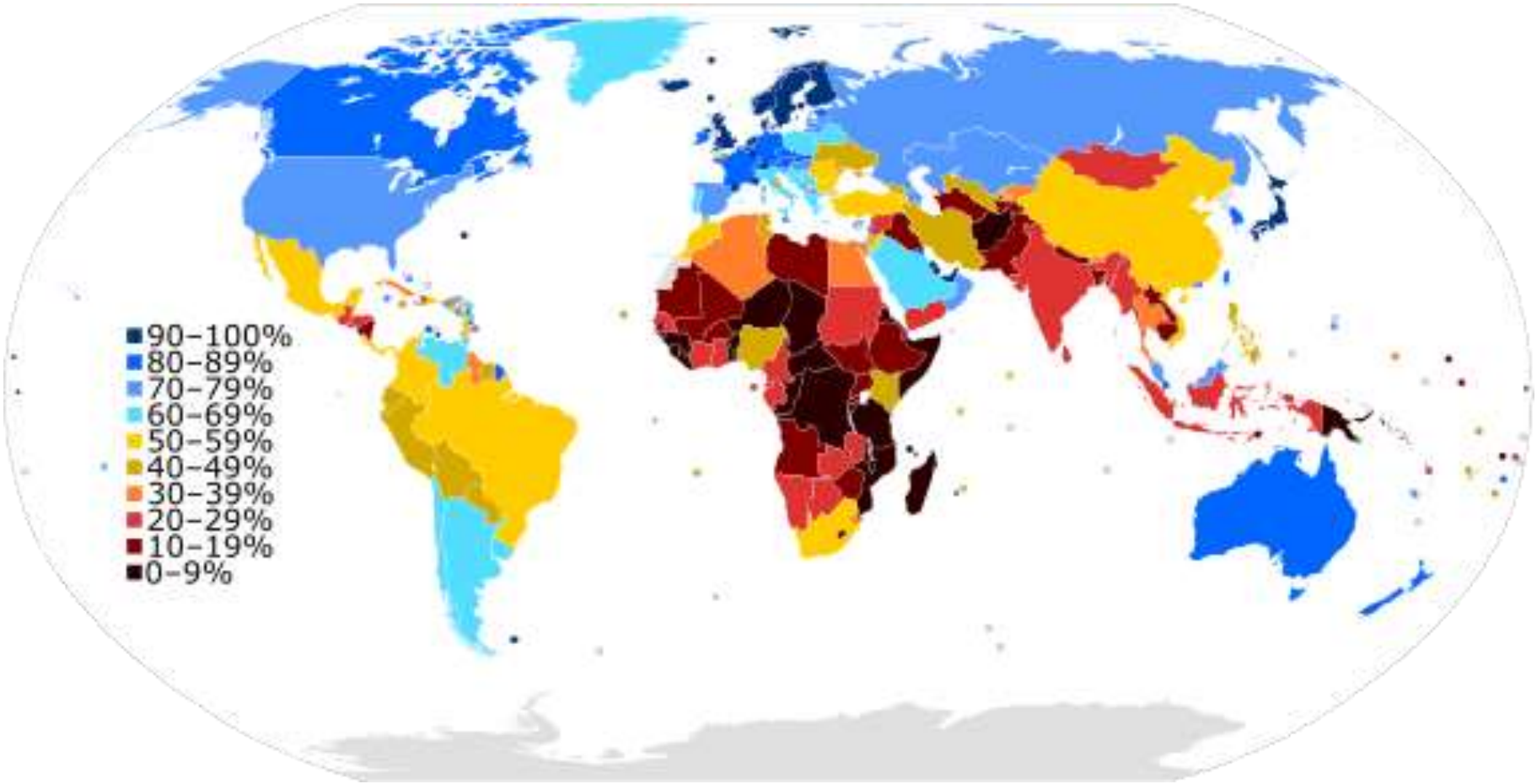
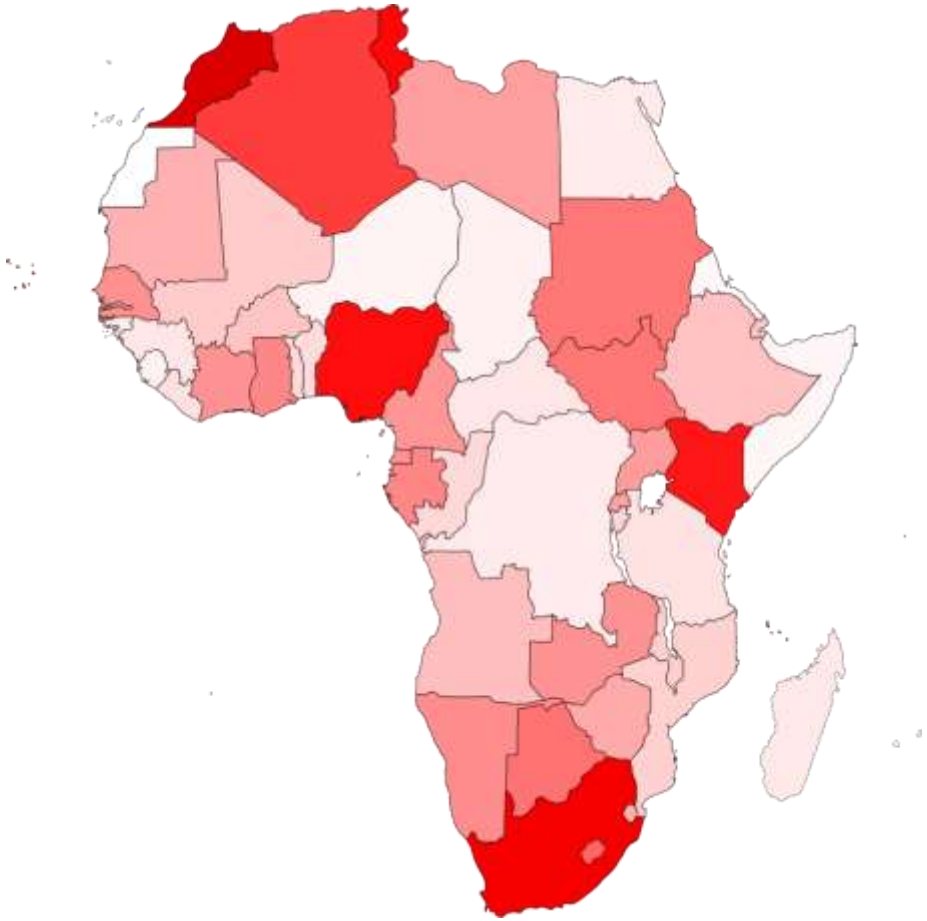
History (cont…)



Internet users in 2015 as a percentage of a country's population
Source: (International Telecommunications Union)

History (cont…)



A map of the percentage of the population of African countries using the Internet in 2015
Source: (International Telecommunications Union)

History (cont…)

- Most of the countries in Africa were connected to the Internet during 1995 and 1996.
  - Prior to 1995, only four countries namely South Africa, Egypt, Zambia and Tunisia were connected.
  - In 1996 when 25 countries in Africa joined the “Internet village”.
  - In January 1997, Ethiopia became the 26th country in Africa to join the Internet.
  - Currently all countries in Africa have Internet connection.

History (cont…)

- Internet expansion (as of websites):

Month/Year	Number of websites
1/93	5
12/93	623
6/95	23,500
1/96	100,000
1/97	646,160
8/97	1,269,800
Today (2022)	[illegible]

Internet Services

- Internet is best characterized by the services it provides.
- It provides a wide range of capabilities for communicating and retrieving information.
- It offers access to data in the form of text, graphics, sound, software and also people through a variety of services and tools for communication exchange.
- The services provided by the Internet is used for interaction, business, marketing, downloading files, sending data etc. Various Internet services are Electronic Mail, World Wide Web (WWW), File Transfer Protocol (FTP), Chat Rooms, Mailing list, Instant Messaging and News Groups.

World Wide Web (WWW)

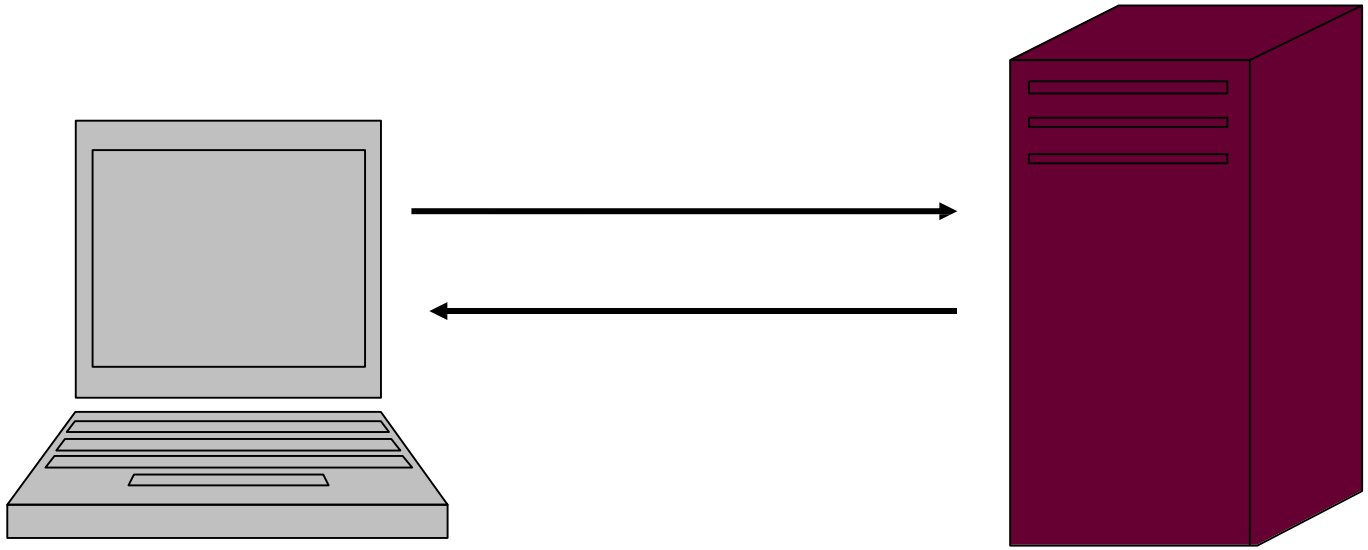
- A collection of interlinked multimedia documents that are stored on the Internet and accessed using a common protocol (HTTP).

World Wide Web (WWW) [Cont…]

- Terms:
  - **Web server** - A Web server is a piece of computer software that can respond to a browser's request for a page, and deliver the page to the Web browser through the Internet
  - **Hypertext** - Machine-readable text that is not sequential but is organized so that related items of information are connected
    - Text that, when clicked, sends you to another piece of text or location
  - **Internet Service Provider (ISP)**
    - A company that provides dial-in or some other type of access to the Internet for a monthly fee
  - **Uniform Resource Locator (URL)**
    - An address on the Internet, such as http://www.bdu.edu.et/scee, which enables computers and other devices to visit it.

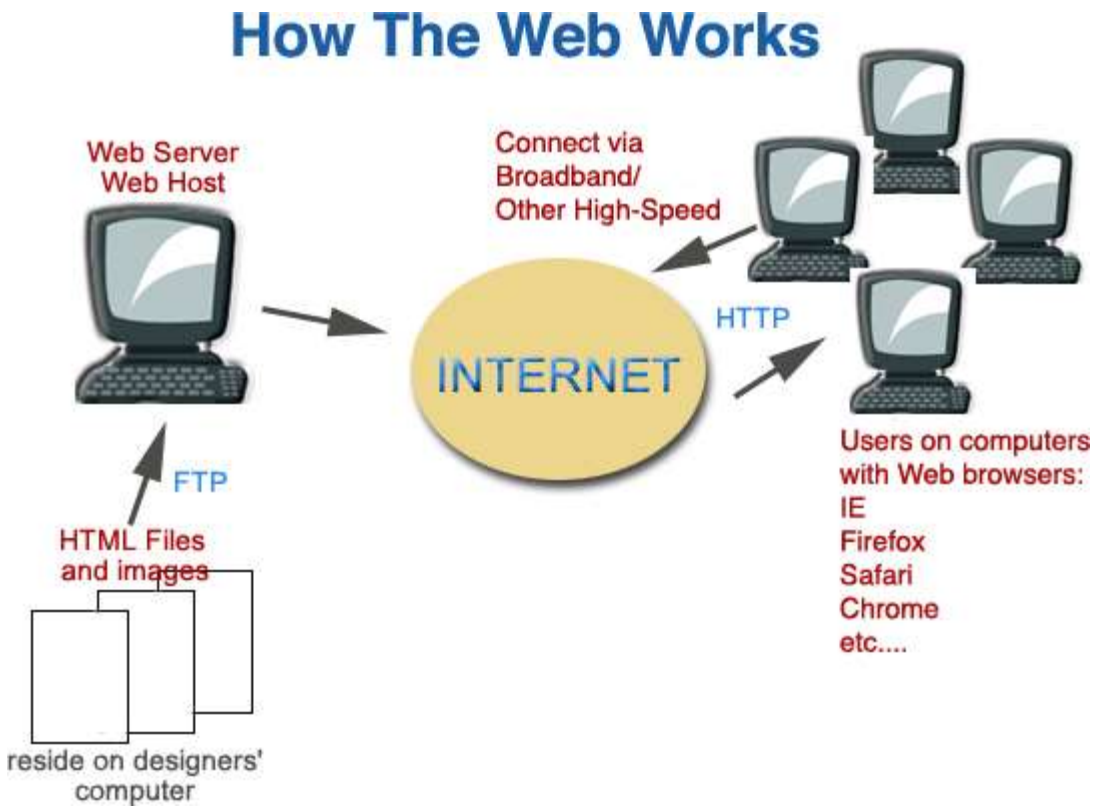
World Wide Web (WWW) [Cont…]

- How does it work?



Simple architecture of the web

World Wide Web (WWW) [Cont…]

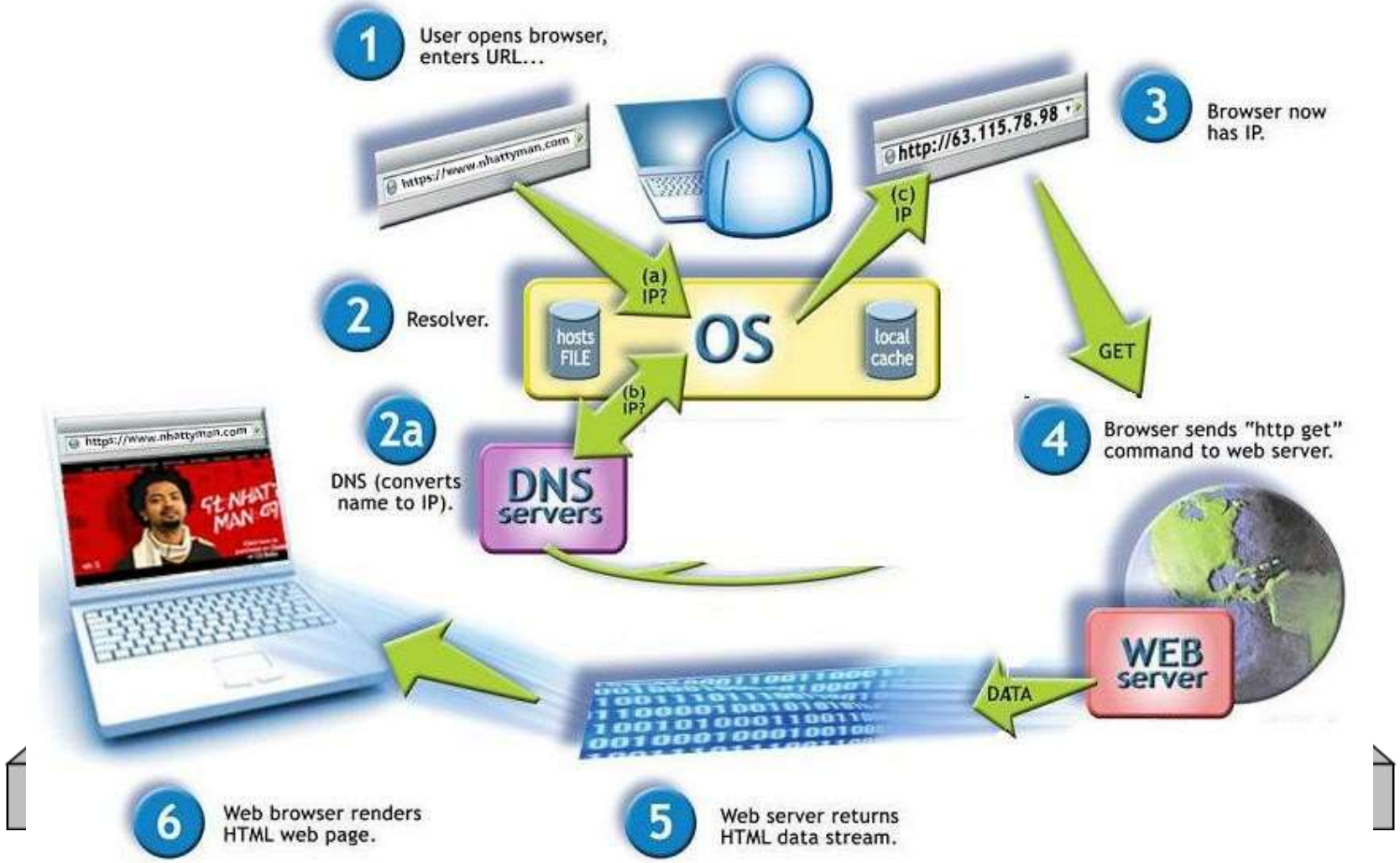


A little bit detailed architecture of the web

World Wide Web (WWW) [Cont…]

- How does it work?
  - Requesting the page https://www.nhattyman.com
  - The browser broke the URL into 3 parts:
    - The protocol ("http")
    - The server name ("www. nhattyman.com")
    - The file name ("index.html")
  - Browser ➔ DNS ……Name to IP translation
  - Browser ➔ Server …..Creation of connection
  - Browser sends a GET request to the server, asking for the file "http:// www. nhattyman.com/index.html".
  - The server then sends the HTML text for the Web page to the browser
  - Browser displays the page based on the HTML tags

World Wide Web (WWW) [Cont…]



Architecture of the web

Showing the *behind-the-scenes* of a user browsing the site https://www.nhattyman.com

World Wide Web (WWW) [Cont…]

- Top Level Domain (TLD) names
  - com - Originally for commercial organizations, but now used by individuals, government agencies, and nonprofits as well. **www.google.com**
  - Net - Internet service providers and other network-related companies. **www.mbc.net, www.mekdim.net**
  - org - Noncommercial (often nonprofit) organizations. **www.ubuntu.org**
  - gov - government agencies. **www.moe.gov.et**
  - mil - military
  - edu - Educational domains. **www.bdu.edu.et**
  - int - International organizations like NATO and the International Red Cross

World Wide Web (WWW) [Cont…]

- More three-or-more-letter TLDs are coming into use, such as:
  - aero - Airlines
  - arpa - Internet infrastructure
  - biz - Businesses
  - coop - Cooperatives
  - info - Anyone
  - museum - Museum
  - name – Individuals

World Wide Web (WWW) [Cont…]

- Internet Address
  - Each computer connected to the Internet must have a unique address, known as IP address.
  - Internet addresses are in the form xxx.xxx.xxx.xxx where xxx must be a number from 0 – 255.

Servers

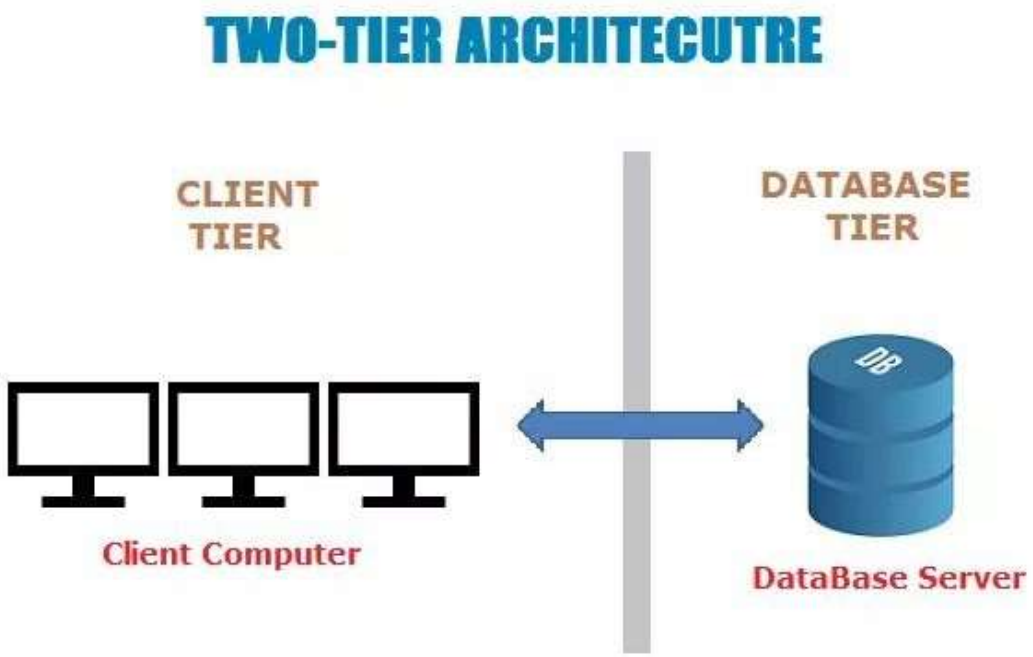
- Web server
  - A computer program that is responsible for accepting HTTP requests from clients (user agents such as web browsers), and serving them HTTP responses along with optional data contents, which usually are web pages such as HTML documents and linked objects (images, etc.).
  - A computer that runs a computer program as described above.
- Common examples of web server software are
  - Apache HTTP server
  - MS Internet Information Server (IIS)
  - Sun ONE web server
  - WAMP (apache+ php+ mysql)

Client-Server Architecture

- A network architecture in which each computer or process on the network is either a client or a server.
- Components of client/server architecture:
  - Communication network
  - Clients
  - Servers

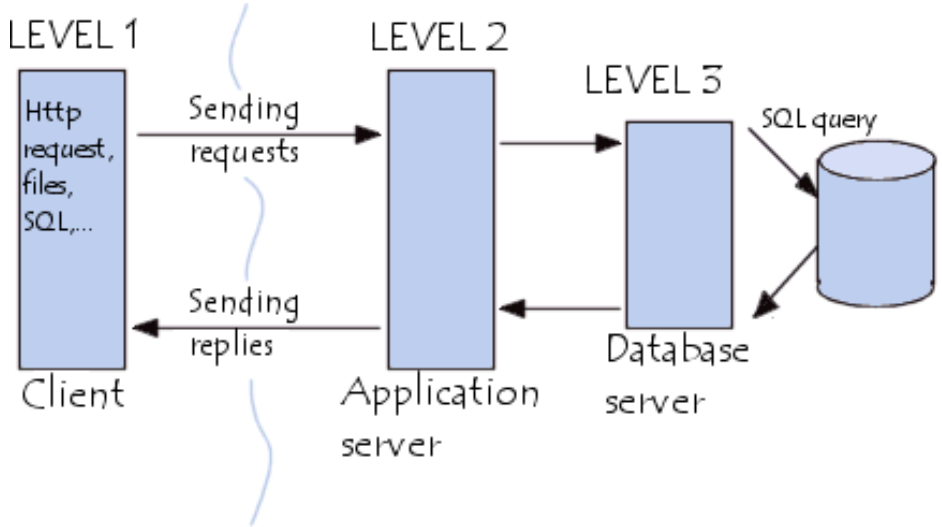
Client-Server Architecture (cont…)

- **2-tier client-server architecture:**
  - A two-tier architecture is a software architecture in which a presentation layer or interface runs on a client, and a data layer or data structure gets stored on a server. Separating these two components into different locations represents a two-tier architecture, as opposed to a single-tier architecture.



Client-Server Architecture (cont…)

- **3-tier client-server architecture:**
  - A three-tier client/server is a type of multi-tier computing architecture in which an entire application is distributed across three different computing layers or tiers. It divides the presentation, application logic and data processing layers across client and server devices.
  - A three-tier client/server adds an additional layer/tier to the client/server-based two-tier models. This additional layer is a server tier that acts as an intermediary or middleware appliance.



URL

- URL (Uniform Resource Locator)

  Example: *http://cs.calstatela.edu:8080/cs320stu31/index.html*

  protocol domain port file

HTTP

- Hypertext Transfer Protocol (HTTP):
  - The Hypertext Transfer Protocol (HTTP) is an application protocol for distributed, collaborative, hypermedia information systems. HTTP is the foundation of data communication for the World Wide Web, where hypertext documents include hyperlinks to other resources that the user can easily access.
  - HTTP was developed to facilitate hypertext and the World Wide Web.

HTTP (cont…)

- HTTP/1.0 allowed only connectionless message passing
  - each request/response required a new connection
  - to download a page with images required multiple connections
  - can overload the server, require lots of overhead
- HTTP/1.1 provides persistent connection by default
  - once client & server connect, remains open until told to close it (or timeout)
  - reduces number of connections, saves overhead
  - client can send multiple requests without waiting for responses
  - e.g., can request all images in a page at once
- HTTP/2.0 was published in May 2015.

HTTP (cont...)

- HTTP Request header fields:
  - A request line (e.g., GET /images/logo.png HTTP/1.1, which requests a resource called /images/logo.png from the server).
  - Request header fields (e.g., Accept-Language: en).
  - the client can specify additional information in the request like:
    - **Referrer** - tells server where the user came from useful for logging and customer tracking.
    - **From** - contains email address of user generally not used for privacy reasons.
    - **Authorization** - can send username & password used with documents that require authorization.
    - **If-Modified-Since** - only send document if newer than specified date used for caching.

HTTP (cont…)

- HTTP Response header fields:
  - The first line of the server's response contains a status code.
    - 200 OK — request was processed successfully
    - 301 Moved permanently — document has been moved
    - 304 Not modified — if cached version is up-to-date

    - 400 Bad request — syntax error in client's request
    - 403 Forbidden — client is not allowed access (e.g., protected)
    - 404 Not found — file could not be found

    - 500 Internal server error — server failed
    - 503 Service unavailable — server is overloaded

HTTP (cont...)

- HTTP Response header fields (cont...)
  - In addition to the status code, the server's response may include:
    - Date, Server info, Last-modified, Content-length, Content-type , Expires
  - Example:

```
HTTP/1.1 200 OK
Date: Mon, 23 May 2005 22:38:34 GMT
Content-Type: text/html; charset=UTF-8
Content-Length: 138
Last-Modified: Wed, 08 Jan 2003 23:11:55 GMT
Server: Apache/1.3.3.7 (Unix) (Red-Hat/Linux)
ETag: "3f80f-1b6-3e1cb03b"
Accept-Ranges: bytes
Connection: close

<html>
<head>
  <title>An Example Page</title>
</head>
<body>
  Hello World, this is a very simple HTML document.
</body>
</html>
```

Caching

- Browsers cache pages to save downloading
- Maintain temporary storage (cache) for recent pages when a page is requested, check to see if already in cache if not in the cache, issue GET request
  - when response message arrives, display page and store in cache (along with header info)
- if already stored in the cache, send GET request with If-Modified-Since header set to the data of the cached page
  - when response message arrives,
    - if status code 200, then display and store in cache
    - if status code 304, then display cached version instead

Cookies

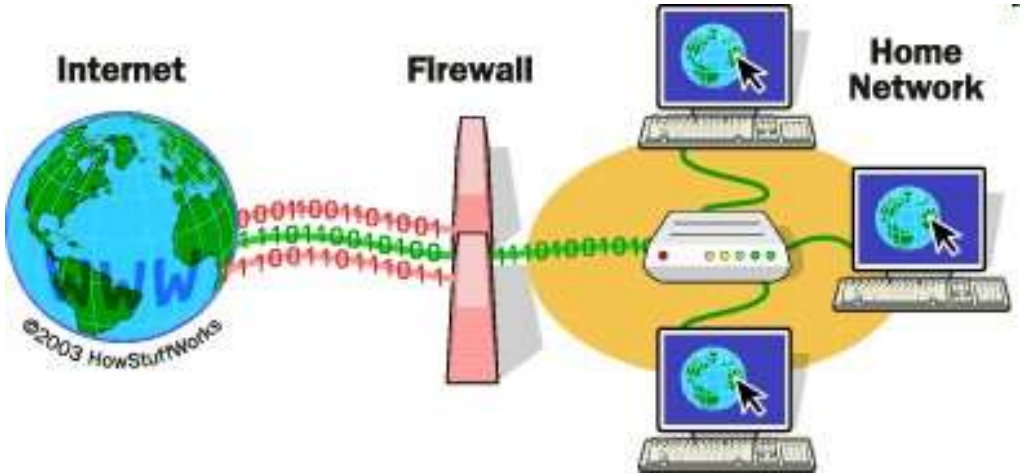
- A cookie is a collection of information about the user.
- Server can download a cookie to the client's machine using the "Set-cookie" header in a response
  - many e-commerce apps require persistent memory of customer interactions

    e.g., amazon.com

    remembers your name, credit card, past purchases, interests.

Firewall

- A firewall is simply a program or hardware device that filters the information coming through the Internet connection into your private network or computer system.
- If an incoming packet of information is flagged by the filters, it is not allowed to pass through.
- Methods
  - Packet filtering
  - Proxy service
  - Stateful inspection



Types of web pages

- Static vs. dynamic pages
  - Static web pages (most web pages are static)
    - contents (text/links/images) are the same each time it is accessed

    e.g., online documents, most homepages
  - Dynamic web pages
    - as the Web moves towards online services and e-commerce, Web pages must also provide dynamic content
    - pages must be fluid, changeable
    - must be able to react to the user's actions, request and process info, tailor services

    e.g., amazon.com, www.facebook.com

Client-Side Scripting

- Used to define the operations performed on the client's machine. Required skills: XHTML, JavaScript, Java, Dreamweaver, Flash, SMIL, XML
- recall: HTML is good for developing static pages
  - can specify text/image layout, presentation, links, …
  - Web page looks the same each time it is accessed
  - in order to develop interactive/reactive pages, must integrate programming written in a separate programming language

Client-Side Scripting (cont...)

e.g., JavaScript, JScript, VBScript programs are embedded in the HTML of a Web page, with tags to identify the program component

e.g., <script type="text/javascript"> … </script>

- the browser executes the program as it loads the page, integrating the dynamic output of the program with the static content of HTML

Client-Side Scripting (cont...)

- Adding dynamic elements to the Web
  - HTML and XHTML pages are static -no real interactivity with the user.
  - Web programming languages such as JavaScript can be used to add interactivity to HTML/XHTML.
  - Examples include responding to mouse and keyboard actions, checking user input, building web pages dynamically, inputting and displaying data, etc.

Client-Side Scripting (cont...)

- Common scripting tasks:
  - Adding dynamic features to Web pages
    - validation of form data
    - image rollovers
    - time-sensitive or random page elements
    - handling cookies
  - Defining programs with Web interfaces
    - Utilize buttons, textboxes, clickable images, prompts, frames

Client-Side Scripting (cont...)

- Limitations of client-side scripting
  - since script code is embedded in the page, viewable to the world
  - for security reasons, scripts are limited in what they can do

    e.g., can't access the client's hard drive
  - since designed to run on any machine platform, scripts do not contain platform specific commands
  - script languages are not full-featured e.g., JavaScript objects are crude, not good for large project development

Server-Side Scripting

- What is server-side code?
  - Software that runs on the server, not the client
  - Receives input from
    - URL parameters
    - HTML form data
    - Cookies: Cookies are files that a site asks the browser to store on the computer of a user to save data to be used in a later session.
    - HTTP headers: define various characteristics of the data that is requested or the data that has been provided
  - Can access server-side databases, e-mail servers, files, mainframes, etc.
  - Dynamically builds a custom HTML response for a client
  - to define the operation to be performed on the server. Skills that are often required: CGI, PHP, ASP, Perl, Java Servlet,…

Server-Side Scripting (cont...)

- To develop a website, three steps:
  1. Obtain the appropriate equipment
     - Web Server – hardware and software
  2. Register the Web Server to an Internet Service Provider (ISP)
     - Obtain the IP address and DNS address
  3. Develop the contents
     - Internet Programming

END